

Proofs

Proofs are the primary means by which mathematical truth is established. The purpose of a proof is to justify the truth of a mathematical statement (called the **conclusion**) by considering other mathematical statements (called **premises**). Once a statement is proved, we refer to it as a **theorem**.

- **Definition 1: Conjunctions ($P \wedge Q$)**

A statement of the form $P \wedge Q$ is true when both P and Q are true. To prove a conjunction, it suffices to prove each of its component statements separately. That is, if we establish P and we also establish Q , we may conclude $P \wedge Q$.

- **Example 1.1**

To prove that a number n is even and greater than 10, we prove separately that n is an even number and that $n > 10$. Combining them yields the result.

- **Definition 2: Disjunctions ($P \vee Q$)**

A statement of the form $P \vee Q$ is true whenever at least one of its components is true. Therefore, to prove a disjunction, it is enough to prove one of its disjuncts (though a false disjunct cannot establish the disjunction).

- **Example 2.1**

To prove that a number n is either even or odd, it suffices to prove that n is even, and the disjunction follows.

- **Definition 3: Indirect Strategy for Disjunction**

Another useful strategy to prove $P \vee Q$ is to proceed indirectly. Assume that one of the disjuncts is false (say $\neg P$) and then prove the other (Q):

$$P \vee Q \equiv \neg P \implies Q.$$

- **Definition 4: Proof by Cases**

If we know that $P \vee Q$ is true, and we can prove R from P , and also prove R from Q , we may conclude R :

- **Example 4.1**

Suppose a natural number is either even or odd, and we want to prove that n^2 is even. We divide the argument into two cases: if n is even, n^2 is even; if n is odd, n^2 is even. Since one of these cases must be true, n^2 is even in either case.

- **Definition 5: Direct Strategy for Implications ($P \implies Q$)**

To prove a statement of the form $P \implies Q$, we must show that whenever P is true, Q is also true.

- **Example 5.1**

To prove that if n is even, then $n + 1$ is odd, we assume that n is even and then show that $n + 1$ is odd.

- **Modus Ponens**

If we establish both $P \implies Q$ and P , we can conclude Q .

- **Definition 6: Indirect Strategy (Proof by Contrapositive)**

Since $P \implies Q \equiv \neg Q \implies \neg P$, we can prove $\neg Q \implies \neg P$ instead of $P \implies Q$.

- **Definition 7: Proof by Contradiction**

To prove $P \implies Q$, suppose that P is true and Q is false ($P \wedge \neg Q$) and show that this assumption leads to a contradiction ($R \wedge \neg R$):

$$P \implies Q \equiv (P \wedge \neg Q) \implies (R \wedge \neg R).$$

We have:

P	Q	R	$\neg Q$	$P \wedge \neg Q$	$R \wedge \neg R$	$P \implies Q$	$(P \wedge \neg Q) \implies (R \wedge \neg R)$
1	1	1	0	0	0	1	1
1	1	0	0	0	0	1	1
1	0	1	1	1	0	0	0
1	0	0	1	1	0	0	0
0	1	1	0	0	0	1	1
0	1	0	0	0	0	1	1
0	0	1	1	0	0	1	1
0	0	0	1	0	0	1	1

Therefore $P \implies Q \equiv (P \wedge \neg Q) \implies (R \wedge \neg R)$.

- **Definition 8: Disproofs and Counterexamples**

A statement is disproved if we show a single instance in which it is false, known as a **counterexample**.

For a universally quantified statement $\forall x P(x)$, a single object x_0 for which $P(x_0)$ is false suffices to disprove the statement.

For an implication $P \implies Q$, a counterexample consists of a situation where P is true and Q is false.

- **Example 8.1**

Consider the statement $x^2 - 1 > 0$ for all $x \in \mathbb{R}$. To disprove it, use the counterexample $x_0 = 1$, which gives $1^2 - 1 = 0 \not> 0$. Thus, the statement is false.

- **Example 8.2**

Prove that the existentially quantified statement $(\exists x \in \mathbb{R}) \left(\frac{1}{x^2+1} > 1 \right)$ is false.

- **Solution 8.2.1**

- We need to show that $\frac{1}{x^2+1} \leq 1$ for every $x \in \mathbb{R}$. Since $0 \leq x^2$ for all $x \in \mathbb{R}$, it follows that $1 \leq x^2 + 1$. Because $x^2 + 1 > 0$, taking the reciprocal reverses the inequality:

$$\frac{1}{x^2 + 1} \leq 1.$$

for every $x \in \mathbb{R}$. Thus, the original statement is false.